

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element G Initial Template

Element G: Construction of a Testable Prototype

Final Prototype Iteration Explanation

Prototype Construction of Design #1 (make as many copies, updating the design number, as needed)

Name: _____

Date: _____

Class: _____



Name: _____

Date: _____

Class: _____

Date: 3/14/25

Details about all steps of construction (add step numbers as needed), including the tools and manufacturing methods used:

	Details	Tools and Manufacturing Methods used
Step 1	We first planned out our design.	Used pencil, paper and a ruler.
Step 2	We then collected the materials to create our design.	Cardboard.
Step 3	We measured the cardboard so we can have the correct measurements for the walls, doors, window, and furniture of the house.	Ruler and pencil
Step 4	We then cut out everything for our furniture and walls of the apartment.	Scissors and exacto knives.
Step 5	We glued together the cardboard walls of the apartment and glued together the pieces of the furniture.	Hot glue gun



Name: _____

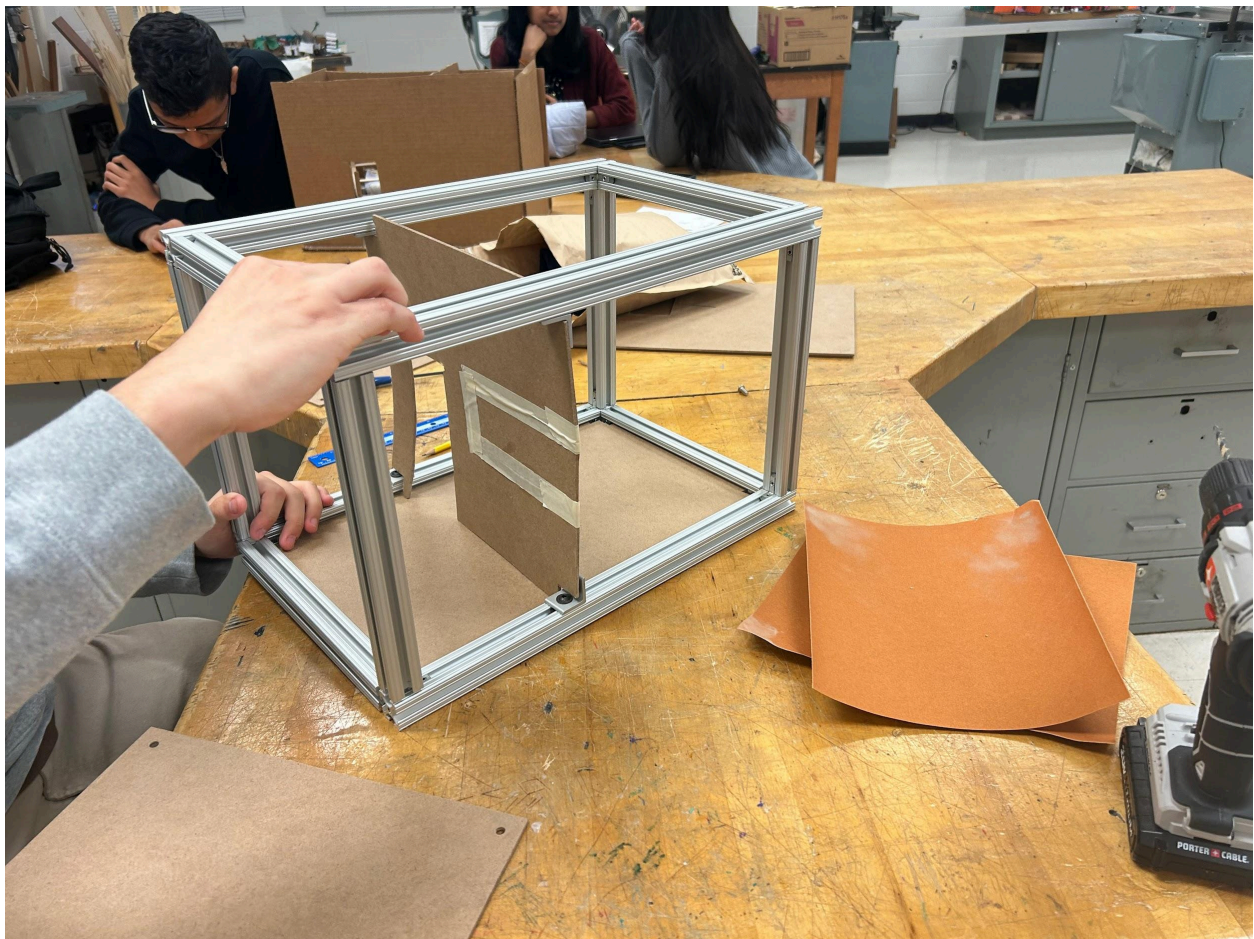
Date: _____

Class: _____

Step 6	We then inserted our furniture into the apartment and our first prototype was complete.	Our hands.
--------	---	------------

Notes of the progression from your blueprint to your (various) physical prototype(s), including why and how aspects of your design may have changed:

Final Prototype Construction of Design #2



Name: _____

Date: _____

Class: _____

Date: 4/9/25

Photos with captions of your full construction process from day 1 of building through to the last day of constructing your final prototype iteration:

Details about all steps of construction (add step numbers as needed), including the tools and manufacturing methods used:

	Details	Tools and Manufacturing Methods used
Step 1	We first collected the materials we needed to create our design and furniture.	Wood, popsicle sticks, cloths, sponges, aluminum bars.
Step 2	We measured and cut out our wood for the windows and doors of our apartments.	Ruler, saw, and Rockwell Shopseries 16-Inch Variable Speed Scroll Saw, RK7315.
Step 3	We inserted aluminum bars on the inside of our design.	Aluminum bars.
Step 4	We completed the code for our detection system on the Arduino cloud	Computer, Arduino Uno, wires, LED, 12v Tone Alarm Ringer Active Buzzer, and a reflective photoelectric sensor module IR Infrared obstacle avoidance.
Step 5	We inserted the walls of the design and screwed them in.	Drill and screws.

Name: _____

Date: _____

Class: _____

Notes of the progression from your blueprint to your (various) physical prototype(s), including why and how aspects of your design may have changed: We could of changed

List of material details, including item name, description, where it was purchased/obtained (include URL if applicable), cost per unit (if applicable), # of units, total cost (if applicable)

Material name	Description	Where material was purchased	Cost per unit	# units	Total cost per item
Hardboard	Wooden walls for the outside of the apartment model	Given from teacher	13.59\$	6	81.54
18 inch 10-series 80/20 Aluminum bars and 10 inch 10-series 80/20 aluminum bars	Bars on the inside of the apartment model to help keep the walls up.	Given from teacher	10.81\$	12	129.72\$
Arduino Uno	The device to download all our code on to for our detection system	Given from teacher	28\$	1	28\$
Buzzer	The item that connects to the arduino	Given from teacher	1\$	1	1\$



Name: _____

Date: _____

Class: _____

	uno to make an alarm sound.				
Photoelectric infrared sensor	Our sensor that detects infrared to alert the detection system	Given from teacher	9.99\$	1	9.99\$
80/20 part 33440 inside-inside hidden corner connector	The item that connects our bars to each other.	Given from teacher	6.83\$	16	109.28\$

Detailed sketches, which might include CAD if you have learned it in your class:

Construction of the Final Prototype (G2)

Design Requirement	How will this prototype allow you to collect objective, measurable data?
Create an successful suppression system	I can light a fire that triggers the suppression system to activate and it can show me how fast and if it suppresses the fire.
Create a successful detection system	I can light a fire to see if it receives the message from the notification system to make an alarm.
Create a successful notification system	I can light a fire to see if it notifies the fire and will send a message to the detection system which will make an alarm.

Name: _____ Date: _____ Class: _____

Create an apartment	I can test it to see if the apartment can stand on its own by putting the apartment on a table to see if the apartment collapses.
Create furniture inside the house	I can test it by determining how comfortable the furniture would be and try to use empathy to see if we would live inside that apartment.
Make the apartment visually appealing	We can test this by looking at our apartment and see if we would be attracted to our apartment if we were someone that has never seen our apartment before.
Finding a safe place to put our detection, notification and suppression system	We can test this by lighting a fire and seeing if any of our systems would catch on fire or see any way of showing that it has been affected by the fire in some way.

Testing or Modeling of Attributes (G3, G4) and Addressing Non-Testable Attributes (G4)

Attribute	Description	Describe your prototype's functionality to test this attribute
The walls of the house	The walls will keep the house up and standing and will truly show that it is an apartment.	This attribute can be tested by seeing if the walls of the house can stand up on their own and make the apartment look like the house.
The notification system	This will be able to detect the fire and send it to the alarm system.	This attribute can be tested by lighting a fire and seeing if the fire is notified or not



Name: _____

Date: _____

Class: _____

The alarm system	This will make an alarming sound to alarm the people of a fire within the home.	This can be tested by lighting a fire and seeing if the alarm gets the notification from the alarm system by making a loud sound from the device
The decorating of the house	This is basically how appealing the house is.	I can test this by looking at my apartment and asking myself if someone would decorate their house like this.
The livability	Basically this is how livable the house is and if the house is made out of livable materials like wood for the walls or cotton for the furniture.	I can test this by making sure every material that was used to build the house would be used in an average apartment house.
Materials	This is to make sure of every material that I need for the apartment I will have.	I can test this by going through the list of the materials and making sure I have every material that I require.
The blueprint	Make sure that the apartment is made off the blueprints my team made for the house.	I can test this by looking at the apartment and looking at the blueprints and seeing if they match.

Addressing Non-Testable Attributes (G4)

Attribute	Justification for why this attribute cannot be modeled or tested with your prototype
-----------	--



Name: _____

Date: _____

Class: _____

Flammability	I cannot test this because I would need to light the house on fire in order to see if it's flammable.
Suppression system	I cannot test this because I will not know if it will work a second time.
Strength of the apartment.	I cannot test this because I would have to burn the house down to test this.
The perspective of someone living in this apartment	I cannot test this because this is not a life sized model and have no clue on how the person living in this house would feel.

Element G: Construction of a Testable Prototype						
	5	4	3	2	1	0
1. Final Prototype Iteration Explanation. The Final Prototype Iteration is _____.	clearly and fully explained	<i>clearly and adequately explained</i>	<i>adequately explained</i>	<i>somewhat explained but with gaps</i>	<i>only minimally explained</i>	<i>is unclear or is missing altogether</i>
2. Construction of the Final Prototype. The Final Prototype Iteration is _____.	constructed with enough detail to assure that objective data on <i>all</i> or nearly all design	constructed with enough detail to assure that objective data on <i>many or nearly all</i>	constructed with enough detail to assure that objective data on <i>some</i> design requirements	constructed with enough detail to assure that objective data on at least a <i>few</i> design requirements could be	<i>not</i> constructed with enough detail to assure that objective data on at least <i>one</i>	<i>not applicable</i> to any design requirements and/or functional claims



Name: _____

Date: _____

Class: _____

	requirements could be determined	design requirements could be determined	could be determined	determined	design requirement could be determined	
3. Testing or Modeling of Attributes. attributes (sub-systems) of the unique solution that can be tested or modeled mathematically are addressed.	<i>All</i>	<i>Most</i>	<i>Some</i>	<i>A Few</i>	<i>No more than one</i>	<i>None of the</i>
4. Addressing non-Testable Attributes. justification is provided for those that cannot be tested or modeled mathematically and thus require expert review.	<i>A well supported</i>	<i>A generally supported</i>	<i>An adequately supported</i>	<i>Developing</i>	<i>Insufficient</i>	<i>No</i>